

Here are the **combined long (16-mark) answers** for the **Cognitive Radio-Based Questions (Q22, Q44, Q45, Q49, Q40)**:

✅ **Q22: Designate the basic cognitive radio (CR) functions within a cognitive cycle.**

Cognitive Radio (CR) operates in a dynamic and intelligent manner through a **cognitive cycle**, enabling efficient spectrum use. The key functions are:

1. **Spectrum Sensing**

- Detects unused frequency bands (spectrum holes).
- Uses techniques like energy detection, matched filtering, and cyclostationary detection.
- Must be accurate and fast to avoid interfering with primary users.

2. **Spectrum Analysis**

- Evaluates the characteristics of detected spectrum holes.
- Considers interference level, channel capacity, and reliability.

3. **Spectrum Decision**

- Selects the most appropriate frequency band based on current QoS requirements and policies.
- Factors include bandwidth availability, latency needs, and mobility.

4. **Spectrum Sharing**

- Coordinates access among multiple secondary users.
- Avoids collisions and interference using MAC-layer protocols and priority scheduling.

5. **Spectrum Mobility**

- Ensures seamless handover when a primary user reclaims the frequency.
- Must guarantee continuity and QoS during transition to a new band.

✅ **Q44: Analyze key requirements and limitations of cognitive-enabled 5G terminals—real-time sensing, decision-making, and energy trade-offs.**

Key Requirements:

1. **Real-Time Spectrum Sensing**

- Terminal must quickly detect and identify available spectrum.
- Critical for ensuring low latency in time-sensitive applications.

2. **Intelligent Decision-Making**

- Embedded AI/ML algorithms to analyze data and decide the best communication channel.
- Must consider context, QoS parameters, and mobility.

3. Adaptive Reconfiguration

- The terminal must switch between bands/interfaces without service interruption.
- Requires software-defined radios (SDR) and dynamic hardware adaptation.

4. Energy Efficiency

- Needs to conserve power despite continuous sensing and computation.
- Sleep modes and energy-aware algorithms are crucial.

Limitations:

- **Processing Overhead:** AI/ML-based cognition demands high computational power.
- **Battery Drain:** Frequent sensing and reconfiguration reduce battery life.
- **Hardware Complexity:** Requires multi-band, multi-mode antennas and reconfigurable RF chains.
- **Latency in Sensing–Decision Loops:** Delayed actions can affect performance in URLLC use cases.

✅ **Q45: Examine how cognitive radio and small-cell networks influence spectrum reuse and interference management in 5G HetNets.**

Cognitive Radio (CR):

- Enables **dynamic spectrum access**, allowing secondary users to exploit underutilized frequencies without interfering with licensed users.
- Enhances **spectrum reuse** by detecting idle bands in real-time.
- Supports **non-intrusive coexistence** through policy-based spectrum etiquette.

Small-Cell Networks:

- Densify the network by deploying low-power base stations (femtocells, picocells).
- Reduce the **transmission range**, which minimizes interference and enables **aggressive frequency reuse**.
- Allow **load balancing**, **offloading** macro-cell traffic, and localized QoS assurance.

In 5G HetNets (Heterogeneous Networks):

- Integration of CR and small cells improves **spectrum efficiency**, **coverage**, and **throughput**.
- Advanced features like **beamforming**, **interference coordination**, and **context-aware resource allocation** are used.

- Result: Better **QoE**, support for URLLC, and network scalability.
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✓ **Q49: Apply cognitive radio techniques to show how dynamic spectrum access enhances 5G network performance. Provide urban/rural use-case examples.**

Dynamic Spectrum Access (DSA) allows unlicensed (secondary) users to utilize licensed spectrum dynamically when not in use.

Performance Enhancements in 5G via DSA:

- Increases **spectrum utilization** and reduces bottlenecks.
- Adapts to **time-varying spectrum availability** and user demands.
- Supports **load-aware, adaptive communications**.
- Enables **opportunistic access** to legacy spectrum (TV white spaces, underused cellular bands).

Urban Use Case:

- In smart cities, CR-enabled IoT devices dynamically use available spectrum for real-time traffic or pollution monitoring.
- Cognitive radios detect and utilize **idle bands** during non-peak hours to avoid congested networks.

Rural Use Case:

- In low-density areas, underutilized licensed bands can be dynamically assigned to broadband services.
 - CR-based 5G terminals support **e-agriculture, remote healthcare, and rural connectivity** without the need for dense infrastructure.
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✓ **Q40: Why is an accurate and efficient radio resource management strategy essential in cognitive 5G networks?**

Radio Resource Management (RRM) in cognitive 5G networks is crucial for:

1. **Maximizing Spectrum Utilization**
 - RRM dynamically allocates idle or shared spectrum to meet changing traffic demands.
2. **Guaranteeing QoS and Reliability**
 - Manages priorities and bandwidth across diverse 5G use cases (eMBB, URLLC, mMTC).
3. **Reducing Interference**
 - Implements sensing-based control to prevent collisions between primary and secondary users.

4. **Optimizing Energy and Latency Trade-offs**

- Balances sensing accuracy, throughput, and power consumption via AI-driven optimization.

5. **Ensuring Seamless Mobility and Handover**

- Supports fast spectrum handover when a user moves or when primary users reclaim the band.

Without robust RRM, cognitive 5G systems may face:

- Spectrum conflicts,
- QoS degradation,
- Latency spikes,
- Inefficient power usage.

An intelligent, policy-aware, and context-driven RRM strategy is therefore **foundational to the success of cognitive-enabled 5G networks.**